

SRS v2.0 Revision Pack

OmniERP — The Invincible Core
Enterprise-grade ERP with deep ETA compliance and the Trust
Network. AI deferred to post-launch.

Supersedes v1.0 §0.3, §5, §6.1, §6.3–6.6, §9.9, §12, §13

DOCUMENT	SRS v2.0 Revision Pack
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AUDIENCE	Engineering, Product, Investors, AI coding agents
STATUS	Draft — 8 open questions (Q-11 to Q-18)

0. Change Log v1.0 → v2.0

#	CHANGE	IMPACT
C-01	AI Subsystem deferred to post-launch (Phase 7+). \$7 remains architecturally valid but is descoped from V1	-8 weeks
C-02	FIFO costing added to V1 alongside WAC, with a full cost-layer ledger	+5 weeks
C-03	Serial number tracking added to V1 (was V2)	+3 weeks
C-04	Batch/expiry with full traceability added to V1 (was P1)	+2 weeks
C-05	Approval workflow engine added to V1 (was out of scope under A-01)	+4 weeks
C-06	Multi-branch with consolidation added to V1 (was P1)	+3 weeks
C-07	Advanced landed cost with retroactive adjustment (was basic allocation)	+2 weeks
C-08	Egyptian on-soil hosting — self-managed Postgres HA, backup, object storage, observability	+3 weeks
C-09	COD carrier settlement reconciliation added (Bosta/Mylerz remittance) — my addition, see §6.14	+2 weeks
C-10	Brand placeholder set to OmniERP	—
C-11	ARQ/Dramatiq confirmed over Celery	—
Net		+16 weeks

0.1 The Scope Reality — Read This

Appendix A of v1.0 argued for launching earlier by deferring the hardest subsystems. You accepted the AI deferral and then replaced it with more than twice as much enterprise depth. That may well be the right call — "NetSuite-grade core, no AI" is a coherent and defensible position, and it's a better position than "shallow core, impressive AI". But it is not the *earlier launch* that Appendix A was arguing for.

Two engineers. ~56 weeks. Three genuinely hard subsystems (costing engine, ETA compliance, approval + multi-branch), each of which a mid-size ERP vendor would staff with a dedicated team.

I have therefore split every enterprise requirement into:

- **Tier A — launch-blocking.** Without this the product is not credible to the target segment.
- **Tier B — fast-follow.** Ships within 90 days of launch. Real, promised, scheduled — just not gating revenue.

My recommendation: launch on Tier A. Tier A alone is already deeper than Daftra, Qoyod, or Wafeq, and comparable to a well-configured Odoo. Tier B is what takes you to NetSuite comparison territory, and it is worth far more when it's built against feedback from paying merchants than against our assumptions today.

If you build A+B before launch, add ~14 weeks and plan for a Q4 2027 launch.

1. Revised Assumptions Register (supersedes §0.3)

#	TOPIC	V2.0 DECISION
A-01	Company size	Revised: 1–50 users. Approval workflows required. Segregation of duties required
A-02	Screen operator	Unchanged — clerk/owner-operator on the surface, accountant-grade underneath
A-03	Pricing	Unchanged. Note: enterprise features justify a higher tier; add an "Enterprise" plan gating multi-branch, approvals, and serial tracking
A-04	Entry price	Revised upward. Entry EGP 700–1,200. Professional EGP 2,500–4,000. Enterprise EGP 6,000–12,000
A-05	Costing	Revised: FIFO and WAC both, selectable per item. Full cost-layer ledger. See §6.3
A-06	Multi-currency	Unchanged — required
A-07	Landed cost	Revised: advanced. Includes retroactive adjustment after goods issue
A-08	Batch/serial	Revised: both in V1. Full traceability
A-09	Language	Unchanged — Arabic-first, RTL-first
A-10	AI	Revised: fully deferred to post-launch
A-11	Mobile	Unchanged — PWA. Addition: mobile approval flow required (§6.12)
A-12/13	Forecasting / anomaly	Deferred. Exception: rule-based anomaly checks that protect data integrity (negative stock, margin below floor, duplicate bill) remain in V1 as validation rules , not "AI"
A-14	Real-time	Unchanged
A-15	Workers	Confirmed: ARQ or Dramatiq
A-16	Breaking changes	Confirmed: greenfield, zero production tenants. Full freedom
A-17	Team	2 engineers, AI-agent-assisted
A-18	Design partners	Now critical. Recruit 5 before Phase 1 completes; at least 2 must be importers to validate FIFO and landed cost
A-19	Hosting	NEW: Egyptian soil. Self-managed. See §9.13
A-20	ETA signing	NEW: Cloud HSM via authorized service provider. Procurement starts now
A-21	Carriers	NEW: Bosta and Mylerz for V1
A-22	Brand	NEW: OmniERP (placeholder)

2. Revised Scope (supersedes §5)

In Scope — V1 (Tier A)

Core platform · Contacts · **Inventory with FIFO+WAC cost layers, multi-warehouse, serial, batch/expiry** · Sales + POS · **Purchases with advanced landed cost** · **Accounting with multi-branch and dimensions** · **Approval workflow engine** · ETA e-Invoice + e-Receipt · Trust Network · **Bosta/Mylerz integration with COD settlement reconciliation** · Plugin platform with widget dashboard · Arabic/RTL-first UI, light+dark · Command palette · PWA.

In Scope — Tier B (fast-follow, ≤90 days post-launch)

Branch consolidation with eliminations · Budgets vs actual · Fixed asset register · Advanced dimensional reporting · Recurring/accrual/prepayment schedules · Quality inspection and quarantine · Supplier price agreements · Consignment stock.

Deferred to Post-Launch (V2)

The entire AI subsystem — Copilot, OCR, forecasting, ML anomaly detection · Native mobile apps · GCC/ZATCA · Manufacturing/BOM · Payroll · Project accounting · Full CRM · Third-party plugin SDK.

Architected-For, Not Built

ZATCA · Neo4j Trust export · **AI tool layer** (the ~60 Copilot tools of v1.0 §7.3 must be built as the *reporting service layer* now, so the AI phase later is a thin adapter, not a rewrite — see §6.15) · Tenant sharding · Manufacturing hooks in the stock ledger.

3. What "Invincible" Actually Means

Before the requirements, a definition — because "invincible" is usually mistaken for "has more features", and that mistake is how ERPs become slow, buggy, and unmaintainable.

An invincible core is one where **these seven properties hold under concurrency, at scale, forever**:

1. **The ledger always balances.** Every journal entry, every period, every tenant, checked continuously, not on demand.
2. **Stock quantity and stock value always agree** — the inventory GL account balance equals the stock valuation report, to the piastre, on any date.
3. **Nothing is ever lost.** Every business event that commits produces every downstream effect, exactly once, even across crashes.
4. **Nothing is ever silently changed.** Every mutation of a financial record is attributable, timestamped, and reversible-by-reversal rather than by edit.
5. **Concurrency is correct.** Two users invoicing the last unit of stock at the same millisecond produce one sale and one clear failure, not two sales or one corrupted quantity.
6. **The past is immutable.** Closed periods cannot move. Historical costs cannot be retroactively rewritten by today's average.
7. **Every number is explainable.** Any figure in any report drills to the transactions that produced it, in two clicks.

Features are what merchants buy. **These seven properties are what stop them leaving.** Where a feature and one of these properties conflict, the property wins. That rule is the whole of §3.

4. §6.3 REVISED — Inventory & Costing (Enterprise)

This is the hardest section in the document. Read §4.1 before the requirement tables.

4.1 The Costing Architecture

Supporting FIFO and WAC simultaneously, with serials, batches, landed cost, and retroactive adjustment, requires one unified structure: **the cost layer ledger**. Do not implement FIFO and WAC as two separate code paths — that way lies permanent divergence and unreconcilable reports.

Three tables carry the entire costing system:

```
stock_movements      immutable quantity ledger — WHAT moved
  id, item_id, variant_id, warehouse_id, branch_id,
  batch_id?, serial_id?, qty (signed), movement_type,
  source_doc_type, source_doc_id, occurred_at, posted_at

cost_layers          immutable value ledger — WHAT IT COST
  id, item_id, variant_id, warehouse_id,
  batch_id?, serial_id?,
  inbound_movement_id, sequence_no,
  qty_received, qty_remaining,
  unit_cost_original, unit_cost_current,  <- landed cost adjusts this
  landed_cost_applied, currency, fx_rate,
  received_at, is_provisional

cost_consumptions    the link — WHICH LAYER PAID FOR WHICH ISSUE
  id, outbound_movement_id, cost_layer_id,
  qty_consumed, unit_cost_at_consumption,
  extended_cost, is_provisional,
  adjustment_of_id?  <- for retroactive true-ups
```

How each method uses it:

METHOD	LAYER BEHAVIOUR	CONSUMPTION RULE
FIFO	One layer per inbound receipt	Consume layers by <code>sequence_no</code> ascending until qty satisfied. May span multiple layers
WAC	One open layer per (item, warehouse) ; each receipt merges into it, recomputing <code>unit_cost_current</code>	Consume from the single layer at current average
Specific (serial)	One layer per serial unit, <code>qty_received = 1</code>	Consume the exact layer for the exact serial
FEFO (batch picking)	FIFO layers, but <i>suggested</i> picking order is by expiry	Cost still FIFO; picking order is FEFO

This is the single design decision that makes the rest of the module tractable. Get it right in Phase 1 and everything else follows; get it wrong and you will rewrite inventory in year two.

4.2 The Four Costing Problems Everyone Gets Wrong

Problem 1 — Retroactive landed cost. Goods arrive in January and are sold. The freight invoice arrives in March. The COGS already posted in January is now wrong.

Solution (FR-330): landed cost applied to a layer generates a **cost adjustment**, not a rewrite. The system: (a) increases `unit_cost_current` on the remaining layer quantity, (b) for the already-consumed portion, posts a **COGS true-up journal entry dated in the current open period** referencing the original consumptions, (c) records the adjustment lineage in `cost_consumptions.adjustment_of_id`. **Never edit the January entry.** Never restate a closed period. The historical margin report for January is annotated as adjusted, with a link to the adjustment.

Problem 2 — Negative stock under FIFO. You sell 10 units when the ledger shows 6. There are no layers to consume for the remaining 4.

Solution (FR-331): create a **provisional layer** at the item's last known cost (or standard cost), flag `is_provisional = true`, post COGS at that provisional cost, and register the issue in a `pending_cost_reconciliation` queue. When real stock arrives, a worker replaces the provisional cost with actual cost and posts the variance to a **Stock Cost Variance** account. The merchant sees a persistent, dismissible warning listing unreconciled provisional costs. **This queue must be drained to zero before period close — it is a hard close gate.**

Problem 3 — Backdated transactions. A user enters a purchase receipt dated three weeks ago, after ten sales have already consumed later layers. Under FIFO, the entire consumption sequence since that date is now wrong.

Solution (FR-332): **Blocked into closed periods, always.** Within an open period, a backdated inbound movement triggers a **cost recalculation job** for that `(item, warehouse)` from the insertion date forward: replay consumptions in `occurred_at` order, recompute layer assignments, and post a single net adjusting entry for the delta. The item is soft-locked during recalculation. If the recalculation would cross a closed period boundary, the entry is rejected with a clear message. Recalculation jobs are queued per item to prevent concurrent replay.

Problem 4 — Concurrency on the last unit. Two POS terminals sell the last unit simultaneously.

Solution (FR-333): stock consumption acquires a **row-level lock on `stock_levels` for `(item, variant, warehouse)`** (`SELECT ... FOR UPDATE`) inside the transaction that writes the movement and the consumption. Locks are acquired in a **deterministic order** (sorted by item id) across multi-line documents to prevent deadlock. Lock wait timeout is bounded and produces a retryable error, not a hang. **This must be load-tested explicitly** with concurrent writers against the same SKU — it is the most common source of silent stock corruption in ERP systems, and it will not show up in normal QA.

4.3 Inventory Requirements

ID	REQUIREMENT	TIER
Stock ledger		
FR-301	Immutable <code>stock_movements</code> ledger. On-hand is derived; <code>stock_levels</code> is a transactionally-maintained cache holding <code>qty_on_hand</code> , <code>qty_reserved</code> , <code>qty_in_transit</code> , <code>qty_available</code>	A
FR-302	Multi-warehouse, multi-branch, with warehouse types (main, retail, transit, quarantine, damaged, consignment)	A
FR-303	Warehouse transfers with a genuine in-transit state and an in-transit GL account. Dispatch and receipt are separate postings; shrinkage in transit is detectable	A
FR-304	Bin/location within warehouse, with directed put-away and picking	B
Costing		

ID	REQUIREMENT	TIER
FR-320	Cost layer ledger per §4.1. cost_layers and cost_consumptions are append-only	A
FR-321	Costing method configurable per item, defaulting from item category, defaulting from tenant. Immutable once the item has any movement — changing method requires a documented revaluation procedure	A
FR-322	FIFO with multi-layer consumption across a single issue	A
FR-323	WAC as a single-open-layer specialisation of the same engine	A
FR-324	Specific identification for serialised items	A
FR-325	Standard cost with purchase price variance	B
FR-326	Manual stock revaluation with mandatory reason and GL posting	A
FR-327	Costed stock ledger report — every movement with qty, unit cost, extended cost, and running balance in both quantity and value. This is the report an auditor asks for	A
FR-328	Stock valuation as-of any date, reconciling to the inventory GL account to the piastre. Any variance is an alertable incident, not a rounding note	A
FR-329	Cost layer inspector UI — for any item, show open layers, their age, and their cost. Merchants distrust costing they cannot see	A
FR-330	Retroactive landed cost adjustment per Problem 1	A
FR-331	Provisional costing and reconciliation queue per Problem 2	A
FR-332	Backdated transaction recalculation per Problem 3	A
FR-333	Concurrency-safe consumption per Problem 4	A
Serial numbers		
FR-340	Serialised items flagged per item. Serial is a first-class entity , not a text field: <code>serials(id, item_id, variant_id, serial_no, status, current_warehouse_id, current_owner_contact_id, cost_layer_id, warranty_start, warranty_end, received_doc_id, issued_doc_id)</code>	A
FR-341	Serial lifecycle: <code>in_stock</code> → <code>reserved</code> → <code>sold</code> → <code>returned</code> → <code>in_stock</code> / → <code>scrapped</code> / → <code>under_repair</code> . Illegal transitions rejected at the database level	A
FR-342	Serial capture is mandatory and validated on receipt and on issue for serialised items. Quantity must equal the serial count — enforced, not warned	A
FR-343	Serial uniqueness enforced per item across the tenant. Duplicate receipt of an in-stock serial is rejected	A
FR-344	Serial genealogy : full history — received on which PO, from which supplier, at what cost, sold on which invoice, to which customer, on what date, returned when, warranty status	A
FR-345	Return validation: a returned serial must have been sold, and to that customer. Blocks the most common retail fraud	A
FR-346	Bulk serial entry: scan sequence, range generation, CSV paste	A

ID	REQUIREMENT	TIER
FR-347	Serial search from anywhere — the command palette resolves a scanned serial to its full history	A
Batch / lot / expiry		
FR-350	Batch-tracked items flagged per item. <code>batches(id, item_id, batch_no, manufacture_date, expiry_date, supplier_batch_ref, status, qty_received)</code>	A
FR-351	Batch capture mandatory on receipt for batch-tracked items; per-batch quantity must reconcile to the line quantity	A
FR-352	FEFO picking suggestion with an override that requires a reason and is logged	A
FR-353	Expiry management: configurable alert horizons (90/60/30/7 days), an expiring-stock dashboard widget, and an expired-stock quarantine action with GL write-off	A
FR-354	Blocked issue of expired batches — hard block, overridable only by a privileged role with a logged reason	A
FR-355	Full traceability, both directions. Backward: "this batch came from which supplier, on which PO, on what date". Forward: "this batch went to which customers, on which invoices" — the recall report. Must return in under 3 seconds	A
FR-356	Batch-level costing: each batch is its own cost layer	A
FR-357	Quality inspection and quarantine workflow on receipt	B
Operations		
FR-360	Stock reservation on sales order / picking, so available ≠ on-hand and two orders cannot promise the same unit	A
FR-361	Stock take: full and cycle counting, blind count, count sheets by zone, multi-counter, variance report requiring approval before posting, adjustment to a Stock Variance GL account	A
FR-362	Reorder point and reorder quantity per item per warehouse, with a Needs-Reordering view and one-click PO draft	A
FR-363	Item variants (matrix), each with own SKU, barcode, price, stock, serial/batch config	A
FR-364	Multiple units of measure per item with conversion factors (buy in cartons, sell in pieces, stock in pieces)	A
FR-365	Multiple barcodes per item/variant/UoM	A
FR-366	Price lists: multiple, per-contact, quantity breaks, date-bounded promotions, currency-specific	A
FR-367	Kits/bundles with component depletion	B
FR-368	Consignment stock (held but not owned, or owned but not held)	B
FR-369	EGS/GS1 code per item — ETA prerequisite, with a pre-submission validation gate	A
FR-370	Dead stock and slow-mover report with capital-tied-up figure	A
FR-371	ABC classification by value contribution	B

5. §6.12 NEW — Approval Workflow Engine

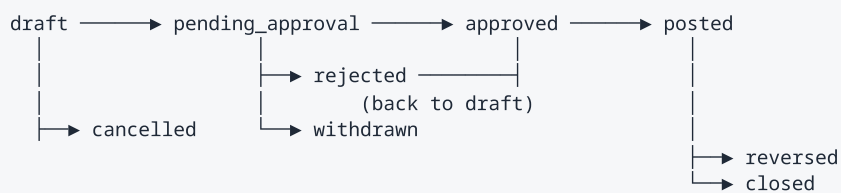
5.1 Design Warning

The instinct when asked for "approval workflows" is to build a generic BPM engine with a visual designer, arbitrary graphs, and scripted conditions. **Do not.** That is a six-month project that will be used by 4% of tenants, will be the source of your worst bugs, and will make every document state transition slow and unpredictable.

Build instead: a fixed, well-understood document state machine plus a declarative rule table. Ninety-five percent of real SMB approval needs are "documents of type X over amount Y need approval from role Z". That is a table, not an engine.

5.2 The Document State Machine

Every approvable document uses the same states. No document type gets custom states.



Rules: - `draft` — freely editable, no GL, no stock effect - `pending_approval` — **locked from editing**, awaiting decision - `approved` — locked, not yet posted (allows batch posting and a cut-off) - `posted` — GL and stock effects committed. **Immutable.** Correction is by reversal only - `reversed` — a reversing document exists and is linked - Documents not subject to any approval rule transition `draft` → `posted` directly. **A tenant with no rules configured must never see an approval UI**

5.3 Requirements

ID	REQUIREMENT	TIER
FR-1201	Uniform state machine across Sales Order, Invoice, Credit Note, PO, Vendor Bill, Payment, Journal Entry, Stock Adjustment, Stock Transfer, Price Change, Credit Limit Change	A
FR-1202	Illegal transitions rejected at the database level (CHECK constraint or trigger), not only in application code. The state machine must hold even if a future worker, script, or bug bypasses the service layer	A
FR-1203	Declarative rule table: (document_type, condition_field, operator, value, approver_type, approver_ref, sequence, is_mandatory) . Conditions limited to: total amount · currency · branch · warehouse · contact · item category · discount percentage · margin percentage · payment terms. Deliberately bounded — no scripting	A
FR-1204	Multiple rules may match; all matching rules apply. Sequential approval by <code>sequence</code> , parallel where sequence is equal	A
FR-1205	Approver resolution: specific user, role, "the creator's manager", or an approval group with any-of / all-of semantics	A
FR-1206	Segregation of duties, enforced. The document creator can never be its approver, regardless of role or permission. A tenant with one user gets no approval rules — the system must refuse to create an unsatisfiable rule rather than deadlock documents	A

ID	REQUIREMENT	TIER
FR-1207	Amount thresholds are evaluated in the tenant's base currency using the document's FX rate. A rule saying "over EGP 50,000" must catch a USD 1,100 purchase	A
FR-1208	Delegation: an approver may delegate to another user for a date range, with the delegation recorded on the approval record	A
FR-1209	Escalation: if a pending approval exceeds N hours, notify and optionally escalate to the next approver	B
FR-1210	Rejection requires a mandatory reason; the document returns to <code>draft</code> with the reason visible to the creator	A
FR-1211	Immutable approval audit: who, when, from what IP, on what device, with what comment, against which rule and which document version hash. Never editable, never deletable	A
FR-1212	Document version hash. The approved content is hashed. If the document is later modified in <code>draft</code> and resubmitted, the prior approval is void — no approving a EGP 5,000 PO and shipping a EGP 500,000 one	A
FR-1213	Approval inbox: a single unified queue across all document types, with filters, bulk approve, and one-click drill into the document	A
FR-1214	Mobile approval flow (PWA) — push notification → summary card → approve/reject with comment. Approvers are owners and managers; they are not at a desk. If approving is inconvenient, documents stall and the whole feature becomes a liability	A
FR-1215	Approval SLA reporting: average time-to-approve by approver and document type. Surfaces the bottleneck human	B
FR-1216	Emergency override by the Owner role, requiring a typed reason, generating a high-visibility audit event and a notification to all managers. Every real business needs this; pretending otherwise means users route around the system entirely	A
FR-1217	Master-data approvals (four-eyes) on: selling price below margin floor, credit limit increase, supplier bank detail change, customer bank detail change. Supplier bank detail change is the single highest-value fraud control in the document	A
FR-1218	Rules are versioned; changing a rule does not retroactively affect in-flight approvals	A

6. §6.13 NEW — Multi-Branch & Organisational Structure

6.1 The Dimensional Model

Branch is **not** a filter bolted onto reports. It is a dimension carried on every transaction from the moment it is created, and it is one of four:

DIMENSION	PURPOSE	REQUIRED
Branch	Legal/operational unit. Owns warehouses, treasuries, numbering series, and a P&L	Mandatory on every transaction
Cost centre	Departmental cost attribution	Optional, configurable as mandatory per account
Project	Job/contract costing	Optional

DIMENSION	PURPOSE	REQUIRED
Salesperson	Attribution and commission	Optional

Design rule: dimensions live on `journal_lines`, not only on document headers. A single invoice may span branches or cost centres at line level. Building this at header level only is a limitation you cannot escape later without a data migration.

6.2 Requirements

ID	REQUIREMENT	TIER
FR-1301	Branch entity with: name, code, address, tax registration, base currency, fiscal calendar reference, active flag	A
FR-1302	Every warehouse, treasury, and bank account belongs to exactly one branch	A
FR-1303	Every transaction carries <code>branch_id</code> , defaulted from the user's branch, overridable by permission	A
FR-1304	Branch-scoped RBAC — a user is assigned to one or more branches and, unless granted cross-branch permission, sees only their branches' data. This must be enforced at the query layer, not the UI layer , and it must be in the cross-tenant isolation test suite alongside tenant isolation	A
FR-1305	Per-branch document numbering series	A
FR-1306	Inter-branch transfers post through an inter-branch clearing account, so each branch's books balance independently. The clearing account must net to zero across branches — a nightly reconciliation invariant	A
FR-1307	Branch-level P&L, balance sheet, trial balance	A
FR-1308	Consolidated statements across branches , with inter-branch elimination of clearing balances, inter-branch sales, and unrealised profit in inter-branch inventory	B
FR-1309	Branch comparison reporting — revenue, margin, expense per branch, side by side	A
FR-1310	Branch-level stock visibility with a cross-branch availability view ("Cairo is out; Alexandria has 40")	A
FR-1311	Multi-dimensional reporting: any report filterable and groupable by any dimension combination	A
FR-1312	Dimension validation rules — e.g. expense accounts require a cost centre; revenue accounts require a salesperson	B
FR-1313	Budget by branch and cost centre, with budget-vs-actual reporting and variance	B
FR-1314	Multi-company (separate legal entities, separate CoA, consolidated group reporting)	V2

7. §6.6 REVISED — Accounting Core (Enterprise)

ID	REQUIREMENT	TIER
Structure		

ID	REQUIREMENT	TIER
FR-551	Hierarchical Chart of Accounts, unlimited depth, Egyptian-localized template, per-account currency, system-account protection (accounts the engine depends on cannot be deleted or retyped)	A
FR-552	Subledger control accounts. AR and AP control accounts are posted to only by the system; a manual journal entry directly to a control account is blocked . The subledger must reconcile to the control account continuously, and divergence is an alertable incident	A
FR-553	Account-level settings: allow manual entry, require dimension, require contact, reconcilable, currency-restricted	A
FR-554	Multi-currency accounts holding both foreign and base amounts. Store both; never recompute historical base amounts from current rates	A
Posting integrity		
FR-560	Every posting is generated by the outbox pattern, is idempotent, and names its source event and rule	A
FR-561	Hash-chained journal entries. Each entry stores a hash of its content plus the previous entry's hash, forming a tamper-evident chain per tenant per period. Verified nightly. Any break is a P1 incident. Cheap to implement, and it is the difference between claiming the ledger is trustworthy and being able to prove it	A
FR-562	Manual journal entries require description, attachment where configured, and approval where a rule matches	A
FR-563	Reversing entries: one click, dated in an open period, bidirectionally linked to the original	A
FR-564	Recurring journal templates with a schedule	B
FR-565	Accrual and prepayment schedules with automatic periodic release	B
Period control		
FR-570	Accounting periods as first-class entities with states: <code>open</code> → <code>soft_closed</code> → <code>hard_closed</code> . Soft close blocks normal users; hard close blocks everyone including Owner	A
FR-571	Close checklist, enforced as gates. A period cannot hard-close while any of these are unresolved: unposted documents · pending approvals · unreconciled provisional costs (FR-331) · unreconciled bank transactions · stock valuation vs GL variance · unsubmitted ETA documents · unbalanced entries · outbox backlog	A
FR-572	Per-module soft close (e.g. close inventory before AR)	B
FR-573	Year-end close: P&L rollup to retained earnings, balance carry-forward, opening entry generation, prior-year lock	A
FR-574	Reopening a closed period requires Owner role, a typed reason, and generates a permanent, prominent audit record	A
Reporting		
FR-580	Trial balance, general ledger, balance sheet, income statement, cash flow (indirect), as-of any date, any dimension combination, comparative to prior period and prior year	A
FR-581	Every figure drills to source transactions in ≤2 clicks	A
FR-582	AR and AP aging with configurable buckets, by branch, by currency, with a base-currency total	A

ID	REQUIREMENT	TIER
FR-583	Customer and supplier statements (كشف حساب), Arabic, PDF and shareable link	A
FR-584	Accountant Mode — dense traditional Arabic presentation of trial balance, GL, and journal register, plus Excel export in the layout Egyptian accountants expect	A
FR-585	Audit file export — complete transaction listing for a period with all dimensions, for an external auditor	A
FR-586	Budget vs actual with variance and drill-through	B
FR-587	Fixed asset register with depreciation schedules and disposal	B
Treasury & banking		
FR-590	Multiple cash boxes and bank accounts per branch, with per-treasury permissions	A
FR-591	Cash shift management: open with float, close with count, variance posting, blind close (the counter does not see the expected figure)	A
FR-592	Bank reconciliation: CSV/OFX statement import, automatic matching by amount+date+reference, manual match UI, unreconciled report	A
FR-593	Inter-treasury transfers with an in-transit account	A
FR-594	Post-dated cheque register with maturity tracking and status lifecycle	A
FR-595	FX revaluation run at period end, posting unrealised gain/loss, with a reversal in the following period	A
Tax		
FR-596	Multi-rate tax engine: standard, zero-rated, exempt, per-line codes, inclusive/exclusive	A
FR-597	Withholding tax (خصم وإضافة) — computed, tracked, reported, with certificate generation	A
FR-598	VAT return report reconciling to the GL and to ETA-submitted documents	A

8. §6.4/6.5 REVISED — Sales & Purchases (Enterprise Deltas)

Only the changes from v1.0 are listed; all v1.0 requirements in §6.4 and §6.5 remain.

Sales

ID	REQUIREMENT	TIER
FR-420	Full document chain with configurable enforcement: Quotation → Sales Order → Picking → Delivery Note → Invoice → Payment. A tenant may enable only the steps they use	A
FR-421	Credit control : limit check at order and at invoice, aged-debt block, blocked-customer flag. Override requires approval (§6.12)	A

ID	REQUIREMENT	TIER
FR-422	Discount approval matrix — discount above X% triggers approval; margin below floor triggers approval	A
FR-423	Partial delivery and partial invoicing with accurate back-order tracking	A
FR-424	Serial and batch selection at picking, validated against reservation	A
FR-425	Returns validated against the original sale, including serial provenance (FR-345)	A
FR-426	Salesperson attribution and commission calculation with configurable schemes	B
FR-427	COGS captured onto the invoice line at the moment of posting — never recomputed from current cost	A
FR-428	Profitability per invoice, per line, per customer, per item, after returns and after COD carrier fees	A

Purchases

ID	REQUIREMENT	TIER
FR-520	Purchase Requisition → PO → Goods Receipt → Vendor Bill → Payment, with approval gates on requisition and PO	A
FR-521	Three-way match with configurable tolerances (percentage and absolute, per-tenant). Within tolerance: auto-approve. Outside: block and route for approval	A
FR-522	Partial receipt and partial billing against a PO, with open-quantity tracking	A
FR-523	Import shipment lifecycle: PO(s) → LC/advance payment → shipped → in customs → cleared → received. Each stage with its own document, dates, and GL treatment	A
FR-524	Advanced landed cost: unlimited cost lines (freight, insurance, customs duty, clearance, inland transport, bank charges, demurrage), each with its own supplier, currency, FX rate, and tax treatment; allocation basis selectable per cost line (value / weight / volume / quantity / manual)	A
FR-525	Landed cost applied before receipt (estimated) with variance posting on actual, and after receipt (retroactive, FR-330)	A
FR-526	Landed cost variance report: estimated vs actual per shipment, per cost type	A
FR-527	Goods received not invoiced (GRNI) accrual account, with an aging report — a classic audit finding when missing	A
FR-528	Supplier price agreements with validity dates and automatic price application	B
FR-529	Purchase returns and debit notes with serial/batch validation	A
FR-530	Supplier performance: on-time delivery rate, price variance, quality rejection rate	B

9. §6.14 NEW — COD Settlement Reconciliation

This section is my addition and I consider it launch-blocking for the primary persona.

For a COD e-commerce merchant, the carrier collects the cash. The merchant does not have the money when the invoice is marked paid — the carrier does, and remits weekly, net of shipping fees, COD handling fees, and return fees, in a settlement file that never quite matches. **Reconciling that file is the single most painful recurring task in a COD business, and no local ERP does it.**

ID	REQUIREMENT	TIER
FR-1401	COD receivable is posted to a "Cash with Carrier" control account on delivery, not to cash. Cash is recognised only on carrier remittance. Anything else materially misstates the cash position	A
FR-1402	Carrier settlement file import (Bosta, Mylerz) with automatic matching on AWB number	A
FR-1403	Automatic posting of shipping fees, COD collection fees, and return fees to expense accounts, per shipment	A
FR-1404	Exception queue: remitted-but-not-delivered · delivered-but-not-remitted · amount mismatch · unknown AWB. Each with a resolution action	A
FR-1405	Carrier aging report — how much each carrier owes, and for how long. Merchants routinely lose five figures here without noticing	A
FR-1406	True order profitability: revenue – COGS – shipping – COD fee – return shipping cost – restocking	A
FR-1407	Return-in-transit tracking with stock only restored on physical receipt, not on carrier status	A
FR-1408	Every carrier status change feeds the Trust Network (\$6.8)	A

Note on FR-1405: this report alone is a demo that closes deals with the primary persona, and it costs a fraction of what the Trust Network costs to build. It should be in the launch pitch.

10. \$6.15 NEW — The Reporting Service Layer (AI Preparation)

Since AI is deferred but not abandoned, build its foundation now at zero extra cost.

Requirement (FR-1501, Tier A): every report, dashboard widget, and analytical query is implemented as a **typed, parameterized service function** with a declared signature, not as inline query logic in a route handler.

```
@reporting_tool(
  name="get_sales_summary",
  description_ar="ملخص المبيعات لفترة",
  required_permission="report.sales.view",
)
async def get_sales_summary(
  tenant: TenantCtx,
  period: DateRange,
  group_by: Literal["day", "week", "month", "branch", "salesperson", "category"] = "day",
  branch_id: UUID | None = None,
  include_returns: bool = True,
) -> SalesSummaryResult: ...
```

Why this matters now: these functions are the report layer, the dashboard widget data source, the export layer, and the public API — today. When the AI phase arrives, the ~60 Copilot tools of v1.0 \$7.3 are **already built**; the AI work reduces to a model that picks a function and fills parameters. That converts an 8-week AI phase into roughly 3 weeks, and it costs nothing extra today because you have to write these functions regardless.

FR-1502 (Tier A): every such function is permission-checked internally against the calling user, never trusting the caller's context. **FR-1503 (Tier A):** every function has a golden test asserting its output against a fixed dataset. This is the seed of the Copilot regression suite.

11. §9.13 NEW — Egyptian On-Soil Hosting

You chose Option B. It is defensible — it is the strongest possible PDPL posture and a genuine marketing asset in a market that distrusts foreign data handling. But it means **you are now the cloud provider**, and the following stop being someone else's problem.

ID	REQUIREMENT	TIER
INF-101	Self-managed PostgreSQL HA. Primary + synchronous standby + async replica, with automated failover (Patroni + etcd, or repmgr). Documented and <i>drilled</i> failover procedure	A
INF-102	Self-managed PITR. WAL archiving to on-soil object storage (MinIO or provider S3-compatible) via pgBackRest or WAL-G. Continuous archiving targeting RPO ≤5 minutes	A
INF-103	Backup verification job — a nightly automated restore of a random tenant schema into a scratch instance, with a row-count and checksum assertion. A backup that has never been restored is not a backup. This is the requirement most likely to save the company one day	A
INF-104	Off-site backup replication to a second Egyptian facility. Single-datacentre is not a DR strategy	A
INF-105	Self-managed object storage for attachments, ETA payloads, and backups, with versioning and lifecycle rules	A
INF-106	Self-managed observability stack (Prometheus + Grafana + Loki, or equivalent) on-soil, since managed APM is likely off the table	A
INF-107	Network: WAF, DDoS mitigation, TLS termination, rate limiting. CDN caveat: an edge CDN caching static assets outside Egypt is normally acceptable since it holds no personal data — but confirm with counsel, and ensure no API response is ever edge-cached	A
INF-108	Capacity planning: 3× headroom on compute and 5× on storage growth. Provisioning lead time from an Egyptian provider is weeks, not minutes — plan procurement a quarter ahead	A
INF-109	Secrets management on-soil (Vault or equivalent). The Trust Network pepper and ETA signing credentials never leave it	A
INF-110	Documented runbooks: failover, restore, scale-out, certificate rotation, incident response. With two engineers, an undocumented system is a single point of failure wearing a human face	A
INF-111	Staging environment mirroring production topology, including replication and failover	A
INF-112	Datacentre due diligence before signing: power redundancy, cooling, network carriers (plural), physical security, SLA and its actual remedies, and the provider's own incident history. Ask for a tour	A

Cost note: self-managed on-soil is typically cheaper per unit of compute than a hyperscaler and considerably more expensive per unit of engineering time. Budget roughly 15–20% of one engineer's ongoing capacity for infrastructure operations, permanently. That is a real, recurring tax on a two-person team, and it should factor into any hiring decision.

Post-launch note: when the AI phase arrives, GPU availability from Egyptian providers is the binding constraint. Begin that conversation with your provider during Phase 1, not in Phase 7.

12. §12 REVISED — Delivery Roadmap

12.1 Phase Plan

PHASE	CONTENT	DURATION	GATE
P0 — Foundation & Infrastructure	Outbox pattern · idempotent consumers · reconciliation invariants · PgBouncer + <code>search_path</code> fix (ARCH-102) · cross-tenant AND cross-branch isolation test suite · migration orchestrator · <code>shard_id</code> plumbing · document sequences · hash-chained audit log · Egyptian infra: Postgres HA, PITR, backup verification, object storage, observability, staging · CI performance gates	7 wks	Isolation suite green · verified restore drill passed
P1a — Costing Engine	Cost layer ledger · FIFO · WAC · specific-ID · provisional costing · backdated recalculation · concurrency locking · costed stock ledger · valuation-to-GL reconciliation	7 wks	Valuation reconciles to GL to the piastre under a 50k-movement concurrent load test
P1b — Inventory Depth	Multi-warehouse · transfers with in-transit · serial numbers full lifecycle · batch/expiry with traceability · reservations · stock take · variants · UoM · price lists	6 wks	Recall trace returns in <3s on 500k movements
P1c — Accounting & Dimensions	Subledger control accounts · period state machine and close gates · hash chain · multi-branch dimension · inter-branch clearing · treasury · bank reconciliation · FX revaluation · withholding tax · full report set with drill-through · Accountant Mode	7 wks	Close checklist blocks on every listed condition
P1d — Approvals & Documents	Approval state machine (DB-enforced) · rule table · segregation of duties · version hashing · approval inbox · mobile approvals · master-data four-eyes	4 wks	Cannot approve own document by any path
P2 — Purchases & Landed Cost	Requisition → PO → GRN → Bill · three-way match with tolerances · import shipment lifecycle · advanced landed cost · retroactive adjustment · GRNI accrual	5 wks	Retroactive landed cost true-up posts correctly across a closed period boundary
P3 — ETA Compliance	Signing abstraction + cloud HSM provider · durable submission queue · rejection translator · EGS registry · credit/debit notes · e-Receipt with 72h window · QR · ETA Health widget · sandbox onboarding	6 wks	97% first-attempt acceptance in sandbox across 500 documents

PHASE	CONTENT	DURATION	GATE
P4 — Sales, POS & UX	Design system · RTL component library · command palette · DataTable · invoice composer · widget dashboard · onboarding wizard · Excel import · POS with offline queue · PWA · dark mode	8 wks	Performance budgets green in CI · 20-second invoice benchmark met
P5 — Trust Network & Carriers	HMAC pepper · k-anonymity · reciprocity gate · consent flow · dispute mechanism · Bosta + Mylerz integration · COD settlement reconciliation · savings widget	5 wks	Legal sign-off received · settlement file reconciles on real carrier data
P6 — Launch Readiness	Penetration test · load test to 3× · DR drill · legal review · DPA and privacy policy · billing · support runbooks · 5-partner pilot	5 wks	Pilot merchants transacting for 30 days without a P1 incident
TOTAL TO LAUNCH (Tier A)		≈60 wks	
P7 — Tier B	Consolidation with eliminations · budgets · fixed assets · quality/quarantine · recurring & accrual schedules · bins · kits · supplier agreements	10 wks	post-launch
P8 — AI Subsystem	AI service on-soil GPU · tier router · Copilot over the existing reporting service layer · OCR · anomaly · forecasting	5 wks <i>(reduced from 8 by FR-1501)</i>	post-launch

12.2 The Honest Number

≈60 weeks to launch. Roughly 14 months. With two engineers, no illness, no scope creep, no ETA surprises, and no infrastructure incidents.

Apply a standard 1.3× factor for a first-time build of this complexity and you get **17–18 months**. If that is unacceptable, the levers, in order of how much I recommend them:

- Drop serial numbers to Tier B (–3 wks).** Your primary persona sells fashion accessories, not electronics. Serials serve the secondary persona and can arrive at month 15. *Strongest recommendation.*
- Drop multi-branch to Tier B (–3 wks).** A single-branch merchant is the launch ICP. Keep the `branch_id` column and the dimensional model in the data layer from day one — that costs nothing — and defer the consolidation, scoping, and UI.
- Ship WAC only at launch, FIFO in month 15 (–4 wks).** Build the cost-layer architecture regardless; WAC is a specialisation of it. This defers the FIFO consumption logic and backdated recalculation, not the design.
- Defer approvals to Tier B (–4 wks).** Only if your launch ICP is genuinely small merchants. Note this contradicts A-01.

Taking levers 1 and 2 alone gets you to **≈54 weeks** with no architectural compromise whatsoever — because in both cases the data model is built and only the surface features are deferred. **That is the cheapest 6 weeks in this document, and I would take it.**

12.3 Parallelisation

The plan assumes meaningful parallelism between you and your co-developer:

WEEKS	YOU (BACKEND/ARCHITECTURE)	CO-DEVELOPER (FRONTEND/UX)
1–7	P0 foundation + Egyptian infrastructure	Design system, RTL component library, DataTable, command palette
8–21	P1a costing, P1b inventory	Inventory UI, serial/batch capture UX, stock take UI
22–32	P1c accounting, P1d approvals	Accounting UI, Accountant Mode, approval inbox, mobile approvals
33–37	P2 purchases and landed cost	Purchase UI, landed cost allocation UI, import shipment tracker
38–43	P3 ETA subsystem	Invoice composer, POS, onboarding, dashboard
44–48	P5 Trust Network, carriers, COD settlement	Trust badge, COD reconciliation UI, savings widget
49–60	Hardening, load testing, pilot support	Polish, empty states, performance, pilot support

Contract-first remains the enabling discipline. Agree the OpenAPI surface for each phase before either side starts. The reporting service layer (FR-1501) is what makes this work — it gives the frontend a stable, typed contract for every screen.

13. §13 REVISED — Risk Register

#	RISK	IMPACT	LIKELIHOOD	MITIGATION
R-01	Cross-tenant or cross-branch data leak	Catastrophic	Medium	ARCH-102, SEC-101, FR-1304. Isolation suite covers both dimensions
R-02	Silent accounting corruption	Severe	High (exists today)	P0 outbox + invariants + hash chain
R-03	Costing engine defects — wrong COGS, valuation drift, layer corruption	Severe. Wrong profit figures destroy trust irrecoverably and are discovered late	High	Cost-layer ledger, property-based tests, nightly valuation-to-GL invariant, concurrent load testing, cost layer inspector UI
R-04	Trust Network PDPL challenge	Severe	Medium	Hashing, k-anonymity, consent, dispute mechanism, and a lawyer . Egyptian soil materially improves this posture
R-05	ETA integration overrun	High	High	Cloud HSM provider procurement starts now, in parallel with P0
R-06	Timeline overrun — 60 weeks becomes 90	High. Runway risk	High	Tier A/B split. Phase gates with real exit criteria. Take levers 1 and 2
R-07	Self-managed infrastructure incident	High	Medium	HA, verified restores, runbooks, staging parity, and the 15–20% ops capacity allowance
R-08	Accountant blocks adoption	High	High	Accountant Mode, audit file export, hash-chained ledger as a trust argument

#	RISK	IMPACT	LIKELIHOOD	MITIGATION
R-09	Two-person key-person risk	High	Medium	Documentation discipline; no undocumented knowledge in costing, tenancy, or ETA
R-10	Trust Network cold start	High	Certain	Concentrated design-partner cohort, historical backfill on import, single-tenant fallback value, honest UNKNOWN copy
R-11	Approval workflow scope explosion into a BPM engine	Medium	High	\$5.1. Bounded rule table. Reject every request for scripting or a visual designer
R-12	14 months with no revenue and no market feedback	High	High	See below

R-12 — The Risk I Care Most About

Fourteen months is a long time to build against assumptions. The specific danger is not technical failure; it is building a superb FIFO engine for importers who turn out to want something else, and discovering it in month 15.

Three mitigations, all cheap:

1. **Recruit 5 design partners before Phase 1 ends (week 21), including 2 importers.** Not for feedback on a demo — for feedback on the costing model specifically. Show them the cost layer inspector and the landed cost allocation screen and let them tell you it's wrong.
2. **Ship an early-access build after Phase 3 (week ~43)** with core + ETA to a handful of merchants, free. That's roughly 4 months of real-world feedback and real ETA submission data before general launch, and it de-risks the most fragile subsystem.
3. **Charge someone something by month 9.** Even EGP 500/month from one merchant changes how you build. Revenue is information.

14. Open Questions — v2.0

#	QUESTION	BLOCKS
Q-11	Do you accept the 60-week estimate, or do you want levers 1 and 2 applied to reach 54? This is the decision I most want an answer to	Everything
Q-12	Egyptian hosting provider shortlist — who, and have you visited? INF-112	P0
Q-13	ETA cloud HSM provider — shortlist and pricing. Start this week	P3
Q-14	Bosta and Mylerz API access — do you have partner credentials, and do their settlement files include the fields FR-1402 needs?	P5
Q-15	Is PDPL counsel engaged for the Trust Network review?	P5
Q-16	Runway: how many months of funding support a 60-week build?	Everything

#	QUESTION	BLOCKS
Q-17	Are you hiring a third engineer? At this scope it changes the plan materially and is probably the highest-ROI decision available	Everything
Q-18	Confirm serial numbers are genuinely required at launch, or accept lever 1	P1b

Appendix B — What I'd Say If You Were My Co-Founder

The plan is sound. The engineering is well-specified. The costing architecture in §4 is, I think, genuinely better than what most mid-market ERPs run on, and the seven properties in §3 are the right definition of the thing you said you wanted.

My concern is not the plan. It is that you responded to "this is too much for two people" by adding more.

That is a normal and understandable instinct — the scope you've described is what the market *deserves*, and every item you added is individually justifiable. But 60 weeks of pre-revenue building with a two-person team is a bet on your runway and your conviction holding steady for over a year, against a market you haven't sold to yet.

The two cheapest things you can do this month:

1. **Take levers 1 and 2.** Six weeks saved, zero architectural compromise, because the data model is built either way and only the surface is deferred.
2. **Hire or partner for a third engineer.** At this scope it is worth more than any feature in this document. The costing engine alone deserves one person's undivided attention for two months, and right now that person is also building ETA compliance and running the datacentre.

And the one thing I'd do before writing another line of code: **take §4.1 and §9 to two real importers and one real COD seller.** An afternoon each. If the cost layer model and the COD settlement report make them lean forward, you have validated the hardest and most expensive parts of this document for the price of three coffees.

If they don't — you've saved a year.